

GARZE, China

WMO: 561460

Lat: 31.6191N

Lon: 99.9976E

Elev: 3394

StdP: 66.67

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.9	-26.6	0.5	1.7	-24.5	0.6	-0.1	10.1	6.4	8.6	5.3	0.5	20	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	24.6	11.9	23.2	11.6	21.8	11.3	14.0	21.1	13.3	20.1	12.7	19.1	2.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
11.6	13.1	16.4	11.1	12.6	15.6	10.6	12.1	15.2	52.4	21.2	50.1	20.1	48.1	19.4	17.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8.9	7.2	5.7	-17.1	27.4	2.0	1.2	-18.6	28.3	-19.8	29.0	-20.9	29.6	-22.3	30.5
			-18.1	15.3	1.8	1.0	-19.4	16.0	-20.5	16.6	-21.5	17.2	-22.8	17.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-2.2	0.5	3.8	7.6	11.4	14.0	15.3	15.1	13.1	8.0	2.2	-1.7
	DBStd	6.92	3.41	2.98	2.88	2.79	3.04	2.55	2.16	2.24	2.69	3.11	3.09	2.87
	HDD10.0	1622	379	266	194	81	19	2	1	1	6	78	233	364
	HDD18.3	4037	638	499	452	321	214	132	97	101	158	321	483	622
	CDD10.0	632	0	0	1	9	63	121	164	159	98	16	0	0
	CDD18.3	4	0	0	0	0	0	1	2	2	0	0	0	0
	CDH23.3	106	0	0	0	1	12	21	26	35	11	1	0	0
	CDH26.7	3	0	0	0	0	0	1	1	0	0	0	0	0
Wind	WSAvg	1.8	1.7	2.0	2.3	2.3	2.2	1.9	1.7	1.6	1.7	1.6	1.5	1.4
Precipitation	PrecAvg	635	4	6	16	30	70	131	117	97	111	39	8	4
	PrecMax	811	15	16	65	77	141	199	194	190	158	84	45	15
	PrecMin	490	0	0	1	10	20	78	62	17	7	6	0	0
	PrecStd	74	4	5	13	17	30	29	35	36	32	21	8	4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.1	14.0	17.6	21.2	25.1	25.9	25.7	26.2	24.8	22.0	15.4	12.1
		MCWB	0.7	1.2	4.0	6.5	10.1	11.7	13.2	11.9	11.6	9.5	2.9	0.7
	2%	DB	9.5	11.7	14.7	18.5	22.4	23.1	23.8	24.2	22.6	18.7	13.2	9.5
		MCWB	-0.9	0.4	2.6	5.1	8.6	12.0	13.0	12.0	11.2	7.9	2.2	-0.5
	5%	DB	7.3	9.8	12.7	16.4	20.5	21.4	22.2	22.6	20.8	16.5	11.4	7.4
		MCWB	-1.9	-0.4	1.8	4.4	7.8	11.6	12.6	11.9	10.9	7.1	1.5	-1.5
	10%	DB	5.3	7.9	11.0	14.6	18.4	19.7	20.6	20.9	18.9	14.2	9.5	5.4
		MCWB	-2.7	-1.1	1.1	3.9	7.4	11.2	12.3	11.6	10.5	6.0	0.7	-2.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.2	1.8	4.8	7.7	11.7	14.6	15.1	14.5	13.2	10.4	4.2	1.1
		MCDB	12.1	11.8	14.9	17.9	20.5	22.0	22.5	21.6	21.3	19.2	11.5	10.6
	2%	WB	-0.5	0.8	3.3	6.3	10.3	13.4	14.2	13.5	12.3	9.2	3.0	-0.3
		MCDB	8.6	10.3	12.5	15.2	19.2	20.3	21.3	20.5	20.0	17.2	11.0	8.4
	5%	WB	-1.6	0.0	2.4	5.4	9.3	12.6	13.6	12.8	11.6	8.1	2.1	-1.3
		MCDB	6.6	8.6	10.8	13.6	17.5	19.2	20.2	19.6	18.7	14.5	9.8	6.6
	10%	WB	-2.6	-0.9	1.6	4.6	8.4	11.9	12.9	12.2	11.0	7.0	1.2	-2.2
		MCDB	4.9	6.7	9.3	12.4	15.8	18.0	19.0	18.5	17.3	12.3	8.3	4.8
Mean Daily Temperature Range	5% DB	MDBR	15.4	14.5	13.2	12.8	12.1	10.5	10.7	11.3	11.4	12.2	14.8	15.7
		MCDBR	18.4	17.9	16.4	16.1	15.4	13.3	13.9	14.7	14.3	15.4	17.8	18.5
		MCWBR	10.1	9.1	7.2	6.1	5.0	4.6	4.9	4.7	5.0	6.0	8.1	9.9
	5% WB	MCDBR	17.8	16.6	14.4	13.5	12.9	11.1	11.2	11.6	12.3	13.2	15.9	17.4
		MCWBR	10.0	8.5	6.4	5.3	4.7	4.4	4.4	4.2	4.5	5.3	7.4	9.6
Clear-Sky Solar Irradiance	taub	0.224	0.257	0.310	0.328	0.313	0.302	0.307	0.307	0.279	0.251	0.225	0.211	
	taud	2.569	2.482	2.313	2.315	2.442	2.587	2.592	2.571	2.648	2.663	2.644	2.641	
	Ebn at Noon	1034	1016	972	961	971	976	971	971	996	1014	1026	1038	
	Edh at Noon	79	96	124	129	115	99	98	99	88	79	73	70	
All-Sky Solar Radiation	RadAvg	3.83	4.61	5.68	6.52	6.39	5.63	5.70	5.40	4.87	4.33	4.10	3.69	
	RadStd	0.22	0.26	0.32	0.46	0.57	0.48	0.41	0.60	0.36	0.21	0.22	0.20	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.54	N/A	-1.94	N/A	-0.42	-0.38	N/A	-172	+88	+3
(47)	Regional (5 neighbors)	+0.49	+0.61	-1.55	+0.77	N/A	N/A	-76	-162	+117	+7

Nomenclature: See separate page